Nesterov momentum for EF21-MuonUSign and EF21-MuonSign

A separate convergence proof

Working proof note

October 22, 2026

Abstract

We prove convergence for the precise Nesterov filter in the paper's server-side-LMO algorithm. Under its smoothness assumptions, both uplink-only and bidirectional EF21 compression attain an expected squared nuclear-gradient rate $O(T^{-1/2})$, with the sharp step and horizon-dependent momentum. Under the paper's generalized smoothness and local lower bounds, the exact downlink variant attains an expected nuclear-gradient rate $O(T^{-1/4})$, with a constant-within-run radius. All tracking recursions, initialization terms, and sufficient constants are supplied below. This is a mathematical proof note. A subsequent, independent Lyapunov proof of both rates is now kernel checked in `lean_proofs/EF21Muon/Nesterov/` and included in the shared article appendix `ef21_nesterov_convergence.tex`. That appendix supplies the constants matched to Lean; this earlier note retains its original constants and is not itself included by the articles.

1 Exact algorithm and assumptions

Let $f = N^{-1}\sum_{j=1}^{N} f_j$ on a tuple of p matrix blocks $X_i \in \mathbb{R}^{m_i\times n_i}$, where $N, p, m_i, n_i \geq 1$ and $r_i = \min(m_i, n_i)$. Fix $0 < \beta \leq 1$ and put $a = 1-\beta$; the paper's momentum parameter is $\mu = a$. At round $t = 1, \ldots, T$, the algorithm is

$$G_t^{(j)} = \nabla f_j(W_{t-1}) + \varepsilon_t^{(j)}, \qquad M_t^{(j)} = aM_{t-1}^{(j)} + \beta G_t^{(j)}, \tag{1}$$

$$V_t^{(j)} = aM_t^{(j)} + \beta G_t^{(j)}, \qquad g_{t,i}^{(j)} = g_{t-1,i}^{(j)} + C_{D,i}(V_{t,i}^{(j)} - g_{t-1,i}^{(j)}), \tag{2}$$

$$g_{t,i} = \frac{1}{N}\sum_{j=1}^{N} g_{t,i}^{(j)}, \qquad X_{t,i} = X_{t-1,i} - \eta_{t,i}\,\mathrm{polar}(g_{t,i}). \tag{3}$$

The server average is exactly the estimator reconstructed by the paper's message-aggregation recursion. The downlink is either exact, $W_t = X_t$, or EF21-P:

$$W_{t,i} = W_{t-1,i} + C_{P,i}(X_{t,i} - W_{t-1,i}). \tag{4}$$

Initialize $M_0^{(j)} = g_0^{(j)} = g_0 = 0$ and $W_0 = X_0$, with deterministic X_0. We also define the bookkeeping values $V_0^{(j)} = M_{-1}^{(j)} = 0$. These do not introduce a stochastic query before round one.

The oracle is an exact, measurable spectral-ball LMO:

$$\langle z, \mathrm{polar}(z)\rangle = \|z\|_*, \qquad \|\mathrm{polar}(z)\|_{2\to 2} \leq 1, \qquad \mathrm{polar}(0) = 0. \tag{5}$$

Every such selection obeys $\|\mathrm{polar}(z)\|_F \leq \sqrt{r_i}$ on block i. In particular, the partial polar factor used in the article is allowed.

Write $\mathcal{F}_{t-1}$ for the history just before the gradients in round t. The query W_{t-1} and all previous state are measurable with respect to this history. For every worker and block assume

$$\mathbb{E}[\varepsilon_{t,i}^{(j)} \mid \mathcal{F}_{t-1}] = 0, \qquad \mathbb{E}[\|\varepsilon_{t,i}^{(j)}\|_F^2 \mid \mathcal{F}_{t-1}] \leq \sigma_i^2. \tag{6}$$

The paper's conditional nuclear-norm variance bound implies this condition. For the generalized-smooth proof we also use conditional independence of the workers' noises, as assumed in the article. It gives

$$\mathbb{E}[\|\bar{\varepsilon}_{t,i}\|_F^2 \mid \mathcal{F}_{t-1}] \le \sigma_i^2/N, \qquad \bar{\varepsilon}_{t,i} = \frac{1}{N}\sum_{j=1}^{N} \varepsilon_{t,i}^{(j)}. \tag{7}$$

Cross-layer independence is unnecessary.

The compressors are conditionally mean-square contractive in the Frobenius norm:

$$\mathbb{E}[\|C(z) - z\|_F^2 \mid z, \text{prior history}] \le (1-\alpha)\|z\|_F^2, \qquad 0 < \alpha \le 1. \tag{8}$$

The parameters are $\alpha_{D,i}$ on the uplink and $\alpha_{P,i}$ on the compressed downlink. The scaled-sign compressors in the article satisfy this pathwise with $\alpha_{D,i} = 1/(m_i n_i)$ and the same downlink parameter. Resolving zero ties in either allowed way causes no change to the proof. In the smooth theorem worker independence is not needed; (6) suffices.

2 Smooth case: both downlinks

Assume $f \ge f_*$ and, for all tuples X, Y and every block i,

$$\|\nabla_i f(X) - \nabla_i f(Y)\|_* \le L_i \|X_i - Y_i\|_{2\to 2}, \tag{9}$$

$$\|\nabla_i f_j(X) - \nabla_i f_j(Y)\|_* \le L_{i,j} \|X_i - Y_i\|_{2\to 2}. \tag{10}$$

All smoothness constants are nonnegative. These are precisely the article's arbitrary-pair assumptions; their restriction on layer interactions is retained. Put

$$\widetilde{L}_i^2 = \frac{1}{N}\sum_{j=1}^{N} L_{i,j}^2, \qquad F_{0,i} = \frac{1}{N}\sum_{j=1}^{N} \|\nabla_i f_j(X_0)\|_F^2, \qquad \Delta = f(X_0) - f_*. \tag{11}$$

Use the sharp schedule $\eta_{t,i} = \gamma_i \|g_{t,i}\|_*$, with fixed $\gamma_i > 0$ within the run. Define

$$\delta_i = \begin{cases} 0, & \text{exact downlink}, \\ 4/\alpha_{P,i}^2, & \text{EF21-P downlink}, \end{cases} \qquad K_i = 3\widetilde{L}_i^2(1 + 2\delta_i), \tag{12}$$

$$A_i(\beta) = \frac{3}{\beta} + \frac{480\beta}{\alpha_{D,i}^2}, \qquad C_i(\beta) = 3\beta + \left(9 + \frac{360}{\alpha_{D,i}^2}\right)\beta^2, \tag{13}$$

$$B_i(\beta) = \left(\frac{3}{\beta^2} + \frac{480}{\alpha_{D,i}^2}\right) K_i + 3L_i^2\delta_i. \tag{14}$$

Theorem 1 (Smooth finite-time bound and tuning)**.** *Suppose* (1)–(8) *and* (9)–(10) *hold. If, for every* i*,*

$$2L_i\gamma_i \le 1, \qquad 20 r_i^2 \gamma_i^2 B_i(\beta) \le 1, \tag{15}$$

then, with $\bar{\gamma} = p^{-1}\sum_i \gamma_i$ *and* $w_i = \gamma_i/\bar{\gamma}$*,*

$$\frac{1}{T}\sum_{t=1}^{T}\sum_i w_i \mathbb{E}\|\nabla_i f(X_{t-1})\|_*^2 \le \frac{4\Delta}{\bar{\gamma}T} + 10\sum_i w_i r_i \left(\frac{A_i(\beta)F_{0,i}}{T} + C_i(\beta)\sigma_i^2\right). \tag{16}$$

In particular, set

$$Z_i = \left(3 + \frac{480}{\alpha_{D,i}^2}\right) K_i + 3L_i^2\delta_i, \tag{17}$$

$$c_i = \left(1 + 2L_i + \sqrt{20}\, r_i \sqrt{Z_i}\right)^{-1}, \qquad \beta = T^{-1/2}, \qquad \gamma_i = c_i T^{-1/2}. \tag{18}$$

With $\bar{c} = p^{-1}\sum_i c_i$ and $w_i = c_i/\bar{c}$, the left side of (16) is at most $D/\sqrt{T}$, where

$$D = \frac{4\Delta}{\bar{c}} + 10\sum_i w_i r_i \left[\left(3 + \frac{480}{\alpha_{D,i}^2}\right) F_{0,i} + \left(12 + \frac{360}{\alpha_{D,i}^2}\right)\sigma_i^2\right]. \tag{19}$$

A common c_i can instead be chosen below every displayed threshold; then all weights are one.

2.1 Tracking at the actual query

Fix a block, temporarily suppress its index, and let $h_t^{(j)} = \nabla_i f_j(W_{t-1})$ for $t \geq 1$. Define $h_0^{(j)} = h_1^{(j)}$ and

$$F_t = \frac{1}{N}\sum_{j=1}^{N} \mathbb{E}\|M_t^{(j)} - h_t^{(j)}\|_F^2, \qquad D_t = \frac{1}{N}\sum_{j=1}^{N} \mathbb{E}\|h_t^{(j)} - h_{t-1}^{(j)}\|_F^2, \tag{20}$$

$$I_t = \frac{1}{N}\sum_{j=1}^{N} \mathbb{E}\|M_t^{(j)} - M_{t-1}^{(j)}\|_F^2, \qquad H_t = \frac{1}{N}\sum_{j=1}^{N} \mathbb{E}\|g_t^{(j)} - V_t^{(j)}\|_F^2, \tag{21}$$

$$P_t = \mathbb{E}\|W_{t,i} - X_{t,i}\|_F^2, \qquad S_t = \mathbb{E}\|X_{t,i} - X_{t-1,i}\|_F^2. \tag{22}$$

Thus $D_1 = 0$, $I_0 = H_0 = P_0 = 0$, and F_0 is the initial constant already defined. A summation without limits below runs from 1 to T.

Lemma 1 (EMA error and its increments)**.** *For $0 < \beta \leq 1$,*

$$F_t \leq aF_{t-1} + \beta^{-1} D_t + \beta^2 \sigma_i^2, \tag{23}$$

$$\sum_t F_t \leq \frac{F_0}{\beta} + \frac{\sum_t D_t}{\beta^2} + \beta T \sigma_i^2, \tag{24}$$

$$\sum_t I_t \leq 4\beta F_0 + 4\sum_t D_t + 3\beta^2 T\sigma_i^2. \tag{25}$$

Proof. Condition on $\mathcal{F}_{t-1}$. Unbiasedness kills the cross term in

$$M_t^{(j)} - h_t^{(j)} = a[(M_{t-1}^{(j)} - h_{t-1}^{(j)}) + (h_{t-1}^{(j)} - h_t^{(j)})] + \beta\varepsilon_t^{(j)}.$$

For vectors in the Frobenius Hilbert space,

$$a^2\|u+v\|_F^2 \leq a\|u\|_F^2 + (a^2/\beta)\|v\|_F^2 \leq a\|u\|_F^2 + \beta^{-1}\|v\|_F^2.$$

The first inequality also holds at $a = 0$ (its two sides are zero). This proves (23). Sum it and discard the nonpositive terminal term $-aF_T$ to obtain (24). Also, directly from (1),

$$I_t \leq 2\beta^2 F_{t-1} + 2\beta^2 D_t + \beta^2\sigma_i^2.$$

Use $\sum_t F_{t-1} \leq F_0 + \sum_t F_t$ and (24), then $\beta^2 \leq \beta$, $\beta^2 \leq 1$, and $\beta^3 \leq \beta^2$. □

Lemma 2 (The Nesterov increment and the uplink)**.**

$$V_t^{(j)} = (1+a)M_t^{(j)} - aM_{t-1}^{(j)}, \tag{26}$$

$$\sum_t \frac{1}{N}\sum_{j=1}^{N} \mathbb{E}\|V_t^{(j)} - V_{t-1}^{(j)}\|_F^2 \leq 10\sum_t I_t, \tag{27}$$

$$\sum_t H_t \leq \frac{160\beta}{\alpha_{D,i}^2} F_0 + \frac{160}{\alpha_{D,i}^2}\sum_t D_t + \frac{120\beta^2}{\alpha_{D,i}^2} T\sigma_i^2. \tag{28}$$

Proof. Equation (26) follows by eliminating βG_t; it is valid at $t = 0$ with the declared bookkeeping values. Consequently

$$V_t - V_{t-1} = (1 + a)(M_t - M_{t-1}) - a(M_{t-1} - M_{t-2}).$$

The squared norm is at most $8\|M_t - M_{t-1}\|_F^2 + 2\|M_{t-1} - M_{t-2}\|_F^2$. Summation proves (27), since $I_0 = 0$.

For $0 < \alpha \le 1$, Young's inequality gives

$$(1 - \alpha)\|u + v\|_F^2 \le (1 - \alpha/2)\|u\|_F^2 + (2/\alpha)\|v\|_F^2. \tag{29}$$

Apply this to $u = V_{t-1}^{(j)} - g_{t-1}^{(j)}$ and $v = V_t^{(j)} - V_{t-1}^{(j)}$, then use compressor contractivity. It follows that

$$H_t \le (1 - \alpha_{D,i}/2)H_{t-1} + \frac{2}{\alpha_{D,i}} \frac{1}{N} \sum_{j=1}^{N} \mathbb{E}\|V_t^{(j)} - V_{t-1}^{(j)}\|_F^2.$$

As $H_0 = 0$, summing yields $\sum H_t \le 40\alpha_{D,i}^{-2} \sum I_t$. Now use Lemma 1. □

Lemma 3 (Downlink and query drift)**.**

$$\sum_t P_t \le \delta_i \sum_t S_t, \qquad \sum_t D_t \le K_i \sum_t S_t. \tag{30}$$

Proof. For an exact downlink $P_t = 0$. Otherwise (29) applied to (4) gives $P_t \le (1-\alpha_{P,i}/2)P_{t-1} + 2S_t/\alpha_{P,i}$. Summing from $P_0 = 0$ proves the first inequality. For $t \ge 2$, smoothness and $\|z\|_{2\to 2} \le \|z\|_F$ give

$$D_t \le 3\widetilde{L}_i^2(S_{t-1} + P_{t-1} + P_{t-2}),$$

because $W_{t-1} - W_{t-2}$ is the sum of the server displacement and the two broadcast errors. Sum, use $D_1 = P_0 = 0$, and apply the first inequality. This proves the second inequality with (12). □

Lemma 4 (Error of the server estimator)**.** *Let* $e_t = \mathbb{E}\|g_{t,i} - \nabla_i f(X_{t-1})\|_F^2$. *Then*

$$\sum_t e_t \le A_i(\beta)F_{0,i} + B_i(\beta) \sum_t S_t + C_i(\beta)T\sigma_i^2. \tag{31}$$

Proof. At a fixed worker, conditional unbiasedness gives the exact identity

$$\mathbb{E}[\|V_t^{(j)} - h_t^{(j)}\|_F^2 \mid \mathcal{F}_{t-1}] = a^2\mathbb{E}[\|M_t^{(j)} - h_t^{(j)}\|_F^2 \mid \mathcal{F}_{t-1}] + \beta^2(1 + 2a)\mathbb{E}[\|\varepsilon_t^{(j)}\|_F^2 \mid \mathcal{F}_{t-1}]. \tag{32}$$

Indeed the two affine noise coefficients are $\beta(1 + a)$ and β, and $[\beta(1 + a)]^2 - a^2\beta^2 = \beta^2(1 + 2a)$. Jensen's inequality for the worker averages, followed by the squared three-term triangle inequality, therefore yields

$$e_t \le 3H_t + 3F_t + 9\beta^2\sigma_i^2 + 3L_i^2 P_{t-1}. \tag{33}$$

The three terms in the decomposition of the error are

$$g_t - \bar{V}_t, \qquad \bar{V}_t - \nabla_i f(W_{t-1}), \qquad \nabla_i f(W_{t-1}) - \nabla_i f(X_{t-1}).$$

All gradient-noise cancellations above are at the predictable query W_{t-1}; none conditions on X_t, which depends on the current noise. Substitute (24), (28), and (30) in (33). The resulting coefficients are exactly (14). □

2.2 Descent and closure of the bound

Proof of Theorem 1. For any two block matrices h, g, set $b = \|g\|_*$, $z = \|h - g\|_*$, and $u = \|h\|_* \le b + z$. Duality and (5) imply

$$\langle h, \|g\|_* \operatorname{polar}(g)\rangle \ge b^2 - bz \ge \tfrac{1}{4}u^2 + \tfrac{1}{2}b^2 - \tfrac{5}{2}z^2. \tag{34}$$

The last inequality follows from $(b - 3z)^2/4 \ge 0$ and $u \le b + z$. The smooth descent lemma and $L_i\gamma_i \le 1/2$ now give

$$\begin{aligned}\mathbb{E}f(X_t) \le \mathbb{E}f(X_{t-1}) &- \frac{1}{4}\sum_i \gamma_i \mathbb{E}\|\nabla_i f(X_{t-1})\|_*^2 - \frac{1}{4}\sum_i \gamma_i \mathbb{E}\|g_{t,i}\|_*^2 \\ &+ \frac{5}{2}\sum_i \gamma_i r_i e_{t,i}.\end{aligned} \tag{35}$$

Here $\|z\|_*^2 \le r_i\|z\|_F^2$, and the upper model follows by integrating (9) along the actual simultaneous block step. The Frobenius displacement satisfies

$$S_{t,i} \le r_i\gamma_i^2 \mathbb{E}\|g_{t,i}\|_*^2. \tag{36}$$

Sum (35) and apply Lemma 4 and (36). The coefficient of $\gamma_i \sum_t \mathbb{E}\|g_{t,i}\|_*^2$ on the positive error side is at most $(5/2)r_i^2\gamma_i^2 B_i(\beta) \le 1/8$. Absorb it into the negative coefficient $1/4$ and discard the remaining nonpositive estimator term. Lower boundedness gives

$$\frac{1}{4}\sum_{t,i}\gamma_i \mathbb{E}\|\nabla_i f(X_{t-1})\|_*^2 \le \Delta + \frac{5}{2}\sum_i \gamma_i r_i [A_i(\beta)F_{0,i} + C_i(\beta)T\sigma_i^2].$$

This is (16) after division. For (18), $\beta^2 B_i(\beta) \le Z_i$, so (15) holds. Finally, for $T \ge 1$,

$$\frac{A_i(\beta)}{T} \le \frac{3 + 480/\alpha_{D,i}^2}{\sqrt{T}}, \qquad C_i(\beta) \le \frac{12 + 360/\alpha_{D,i}^2}{\sqrt{T}}.$$

These yield (19). □

3 Generalized smoothness: exact downlink

In this section $W_t = X_t$, $\eta_{t,i} = s_i > 0$ is constant within a run, and we replace (9) by the article's condition

$$\|\nabla_i g(X) - \nabla_i g(Y)\|_* \le (L_i^{0,g} + L_i^{1,g}\|\nabla_i g(X)\|_*)\|X_i - Y_i\|_{2\to 2} \tag{37}$$

for every pair X, Y and $g = f, f_j$, with nonnegative constants. Write (L_i^0, L_i^1) for f and $(L_{i,j}^0, L_{i,j}^1)$ for f_j. Assume $f_j \ge f_{j,*}$, and put

$$f_\dagger = \frac{1}{N}\sum_{j=1}^N f_{j,*}, \qquad \Delta_0 = f(X_0) - f_\dagger, \qquad q_i = \sqrt{1 - \alpha_{D,i}}, \qquad \kappa_i = \frac{q_i}{1 - q_i}.$$

The constants introduced below cover zero generalized-smoothness coefficients without any division by them.

3.1 First-moment tracking, including filtered noise

For the remainder of this subsection fix a block and suppress i. Let $h_t^{(j)} = \nabla_i f_j(X_{t-1})$ and again set $h_0 = h_1$. Define the first-moment quantities

$$u_t = \frac{1}{N}\sum_{j=1}^{N} \mathbb{E}\|M_t^{(j)} - h_t^{(j)}\|_F, \qquad d_t^{\mathrm{loc}} = \frac{1}{N}\sum_{j=1}^{N} \mathbb{E}\|h_t^{(j)} - h_{t-1}^{(j)}\|_F,$$

$$v_t = \frac{1}{N}\sum_{j=1}^{N} \mathbb{E}\|g_t^{(j)} - V_t^{(j)}\|_F, \qquad d_t^{\mathrm{glob}} = \mathbb{E}\|\bar{h}_t - \bar{h}_{t-1}\|_F,$$

where $\bar{h}_t = \nabla_i f(X_{t-1})$. Thus $v_0 = 0$ and both drifts vanish at $t = 1$. The initial local error is $u_0 = N^{-1}\sum_j \|\nabla_i f_j(X_0)\|_F$.

Introduce a noise-only EMA and a de-noised global bias:

$$Z_0 = 0, \quad Z_t = aZ_{t-1} + \beta\bar{\varepsilon}_t, \quad B_t = \bar{M}_t - Z_t - \bar{h}_t, \quad b_t = \mathbb{E}\|B_t\|_F. \tag{38}$$

Here $b_0 = \|\nabla_i f(X_0)\|_F$ and $B_t = a(B_{t-1} + \bar{h}_{t-1} - \bar{h}_t)$ exactly.

Lemma 5 (First-moment estimator bound)**.** *For every positive prefix length $k \le T$, with all sums from 1 to k,*

$$\sum_t \mathbb{E}\|g_{t,i} - \nabla_i f(X_{t-1})\|_F \le \frac{b_0}{\beta} + 6\kappa_i u_0 + \frac{1}{\beta}\sum_t d_t^{\mathrm{glob}} + 6\kappa_i \sum_t d_t^{\mathrm{loc}} + 6\kappa_i \beta k\sigma_i + 2\sqrt{\beta}\, k\sigma_i/\sqrt{N}. \tag{39}$$

Proof. The triangle inequality, conditional Cauchy–Schwarz, and (1) give

$$u_t \le a u_{t-1} + a d_t^{\mathrm{loc}} + \beta\sigma_i, \qquad \sum_t u_t \le \frac{u_0}{\beta} + \frac{\sum_t d_t^{\mathrm{loc}}}{\beta} + k\sigma_i.$$

Consequently

$$\sum_t \frac{1}{N}\sum_{j=1}^{N} \mathbb{E}\|M_t^{(j)} - M_{t-1}^{(j)}\|_F \le 2u_0 + 2\sum_t d_t^{\mathrm{loc}} + 2\beta k\sigma_i.$$

By (26), the corresponding sum for $V_t - V_{t-1}$ is at most three times this expression. Compressor contractivity and conditional Jensen give

$$v_t \le q_i \left(v_{t-1} + \frac{1}{N}\sum_{j=1}^{N} \mathbb{E}\|V_t^{(j)} - V_{t-1}^{(j)}\|_F \right).$$

Sum from $v_0 = 0$ to obtain

$$\sum_t v_t \le 6\kappa_i \left(u_0 + \sum_t d_t^{\mathrm{loc}} + \beta k\sigma_i \right). \tag{40}$$

Next, (38) yields $b_t \le a b_{t-1} + a d_t^{\mathrm{glob}}$, whence

$$\sum_t b_t \le b_0/\beta + \beta^{-1}\sum_t d_t^{\mathrm{glob}}.$$

The martingale difference property and (7) yield

$$\mathbb{E}\|Z_t\|_F^2 \le \frac{\beta}{2-\beta}\frac{\sigma_i^2}{N} \le \frac{\beta\sigma_i^2}{N}. \tag{41}$$

The effective averaged target satisfies the exact identity

$$\bar{V}_t - \bar{h}_t = aB_t + aZ_t + \beta\bar{\varepsilon}_t.$$

Hence $\mathbb{E}\|\bar{V}_t - \bar{h}_t\|_F \le b_t + 2\sqrt{\beta}\,\sigma_i/\sqrt{N}$, since $\beta \le \sqrt{\beta}$. Finally $\mathbb{E}\|g_t - \bar{h}_t\|_F \le v_t + \mathbb{E}\|\bar{V}_t - \bar{h}_t\|_F$. Summation and (40) prove (39). □

3.2 Coefficient envelopes from lower boundedness

Choose

$$M = 1 + \max\big(\{L_i^1\}_i \cup \{L_{i,j}^1\}_{i,j}\big), \qquad \Lambda_i = \max_j L_{i,j}^1, \tag{42}$$

$$O = M^{-1} \sum_i L_i^0, \qquad \widetilde{O} = M^{-1} \frac{1}{N} \sum_{j=1}^N \sum_i L_{i,j}^0. \tag{43}$$

For $\Delta(X) = f(X) - f_\dagger \geq 0$, define

$$A_i = L_i^0 + L_i^1 O, \qquad B_i = 2ML_i^1, \tag{44}$$

$$\widetilde{A}_i = \frac{1}{N} \sum_{j=1}^N L_{i,j}^0 + \Lambda_i \widetilde{O}, \qquad \widetilde{B}_i = 2M\Lambda_i. \tag{45}$$

These symbols A_i, B_i are used only in this generalized-smooth section.

Lemma 6 (Coefficient envelopes)**.** *For every X,*

$$L_i^0 + L_i^1 \|\nabla_i f(X)\|_* \leq A_i + B_i \Delta(X), \tag{46}$$

$$\frac{1}{N} \sum_{j=1}^N \left(L_{i,j}^0 + L_{i,j}^1 \|\nabla_i f_j(X)\|_* \right) \leq \widetilde{A}_i + \widetilde{B}_i \Delta(X). \tag{47}$$

Proof. For any g satisfying (37) and $g \geq g_*$, take the simultaneous block step $Y_i = X_i - M^{-1} \operatorname{polar}(\nabla_i g(X))$. Integrating (37) along this segment gives

$$g(Y) \leq g(X) - M^{-1} \sum_i \|\nabla_i g(X)\|_* + \frac{1}{2M^2} \sum_i \left(L_i^{0,g} + L_i^{1,g} \|\nabla_i g(X)\|_* \right).$$

Since $M \geq L_i^{1,g}$ and $g(Y) \geq g_*$,

$$\sum_i \|\nabla_i g(X)\|_* \leq 2M(g(X) - g_*) + M^{-1} \sum_i L_i^{0,g}. \tag{48}$$

Use $g = f$, $g_* = f_\dagger$ to obtain the first envelope. Apply (48) to each f_j, average, and use $N^{-1} \sum_j (f_j(X) - f_{j,*}) = \Delta(X)$. Multiplication by Λ_i proves the second envelope. □

3.3 Finite-time and asymptotic bounds

Define nonnegative constants

$$U_i = s_i^2 \left(\frac{1}{2} + \frac{2\sqrt{r_i}}{\beta} \right), \qquad V_i = 12\kappa_i s_i^2 \sqrt{r_i}, \tag{49}$$

$$\nu = \sum_i \left(12\kappa_i s_i \sqrt{r_i}\, \beta\sigma_i + 4s_i \sqrt{r_i \beta}\, \sigma_i / \sqrt{N} \right), \tag{50}$$

$$C_0 = \Delta_0 + 2 \sum_i s_i \sqrt{r_i} (b_{0,i}/\beta + 6\kappa_i u_{0,i}), \tag{51}$$

$$d = \sum_i (U_i A_i + V_i \widetilde{A}_i) + \nu, \qquad b = \sum_i (U_i B_i + V_i \widetilde{B}_i). \tag{52}$$

Theorem 2 (Generalized-smooth Nesterov convergence)**.** *Suppose (1)–(8), (37), and the local lower bounds hold. Use an exact downlink and a constant radius $\eta_{t,i} = s_i$. Then*

$$\sum_{t=1}^T \sum_i s_i \mathbb{E} \|\nabla_i f(X_{t-1})\|_* \leq (1+b)^T (C_0 + Td). \tag{53}$$

In particular, for any fixed $c_i > 0$, choose

$$\beta = T^{-1/2}, \qquad s_i = c_i T^{-3/4}, \qquad \bar{c} = p^{-1} \sum_i c_i, \qquad w_i = c_i/\bar{c}. \tag{54}$$

There is an explicit constant D_{gen}, independent of T, such that

$$\frac{1}{T} \sum_{t=1}^{T} \sum_i w_i \mathbb{E} \|\nabla_i f(X_{t-1})\|_* \le D_{\text{gen}} T^{-1/4}. \tag{55}$$

In particular the minimum expected weighted gradient norm satisfies the same bound. No bounded-gradient hypothesis is added.

Proof. Put $\Delta_t = \mathbb{E}[f(X_t) - f_\dagger]$. Exact oracle duality gives

$$\langle \nabla_i f(X), \operatorname{polar}(g_i) \rangle \ge \|\nabla_i f(X)\|_* - 2\|g_i - \nabla_i f(X)\|_*.$$

The generalized descent lemma, $\|X_{t,i} - X_{t-1,i}\|_{2\to 2} \le s_i$, and $\|z\|_* \le \sqrt{r_i}\|z\|_F$ imply

$$\begin{aligned} \Delta_t + \sum_i s_i \mathbb{E} \|\nabla_i f(X_{t-1})\|_* \le \Delta_{t-1} + 2 \sum_i s_i \sqrt{r_i} \mathbb{E} \|g_{t,i} - \nabla_i f(X_{t-1})\|_F \\ + \sum_i \frac{s_i^2}{2} \mathbb{E}[L_i^0 + L_i^1 \|\nabla_i f(X_{t-1})\|_*]. \end{aligned} \tag{56}$$

For $t \ge 2$, (37) with the coefficient evaluated at X_{t-2} gives

$$\begin{aligned} d_{t,i}^{\text{glob}} &\le s_i \mathbb{E}[L_i^0 + L_i^1 \|\nabla_i f(X_{t-2})\|_*], \\ d_{t,i}^{\text{loc}} &\le s_i \frac{1}{N} \sum_{j=1}^{N} \mathbb{E}[L_{i,j}^0 + L_{i,j}^1 \|\nabla_i f_j(X_{t-2})\|_*]. \end{aligned}$$

Both drifts vanish at $t = 1$. Thus, for every prefix k, their sums are bounded by the corresponding coefficient sums at $X_0, \dots, X_{k-1}$, multiplied by s_i. Sum (56), substitute Lemma 5, and then Lemma 6. This gives

$$Y_k := \Delta_k + \sum_{t=1}^{k} \sum_i s_i \mathbb{E} \|\nabla_i f(X_{t-1})\|_* \le C_0 + kd + b \sum_{t=0}^{k-1} \Delta_t. \tag{57}$$

All terms in Y_k are nonnegative, so $\Delta_t \le Y_t$ and $Y_0 = \Delta_0 \le C_0$. An elementary induction gives

$$Y_k \le (C_0 + kd)(1 + b)^k.$$

For completeness, at induction step k bound each earlier Y_t by $(C_0 + kd)(1 + b)^t$ and use $1 + b\sum_{t=0}^{k-1}(1 + b)^t = (1 + b)^k$, an identity also valid at $b = 0$. Drop $\Delta_T \ge 0$ to prove (53).

Here are explicit horizon-independent constants for (55):

$$\begin{aligned} \bar{U}_i &= c_i^2(1/2 + 2\sqrt{r_i}), & \bar{V}_i &= 12\kappa_i c_i^2 \sqrt{r_i}, \\ \bar{C}_0 &= \Delta_0 + 2 \sum_i c_i \sqrt{r_i}(b_{0,i} + 6\kappa_i u_{0,i}), \\ \bar{d} &= \sum_i (\bar{U}_i A_i + \bar{V}_i \widetilde{A}_i) + \sum_i \left(12\kappa_i c_i \sqrt{r_i}\sigma_i + 4c_i\sqrt{r_i}\sigma_i/\sqrt{N}\right), \\ \bar{b} &= \sum_i (\bar{U}_i B_i + \bar{V}_i \widetilde{B}_i), & D_{\text{gen}} &= e^{\bar{b}}(\bar{C}_0 + \bar{d})/\bar{c}. \end{aligned}$$

Indeed (54) implies $C_0 \le \bar{C}_0$, $Td \le \bar{d}$, and $Tb \le \bar{b}$ for all $T \ge 1$. In particular $TU_i \le \bar{U}_i$ and $TV_i \le \bar{V}_i$; the two noise powers are $Ts_i\beta = c_i T^{-1/4}$ and $Ts_i\sqrt{\beta} = c_i$. Since $(1 + b)^T \le e^{Tb}$ and $T(p^{-1}\sum_i s_i) = \bar{c}T^{1/4}$, (53) gives (55). □

4 Integrability and scope

All expectations used above are well-defined and finite. In the smooth case, global Lipschitz gradients have at most linear growth. Starting from deterministic finite state, (6), the finite-dimensional norm comparisons, and conditional compressor contractivity inductively give square-integrable state at every finite round. Smooth f has at most quadratic growth, so its values are integrable. In the generalized-smooth case the normalized steps put every finite-horizon trajectory in a fixed compact set, independently of the sampled gradients. Continuity bounds the objectives and gradients there; (6) and contractivity then give the remaining integrability. The exact partial polar and scaled-sign selections are measurable.

The result concerns the paper's Nesterov *filter* $V_t = \mu M_t + (1-\mu)G_t$ at the same queried model. It does not change the query to a classical extrapolated Nesterov point. The algorithm order, zero initialization, and server aggregation are those of `aaai_article/signmuon_appendix.tex`, Algorithm `alg:fed_serverlmo`.

Theorems 1 and 2 close the Nesterov-filter extension of the article's two theoretical rates. Their constants are conservative and are not asserted to match the EMA constants in the article or in *Error Feedback for Muon and Friends* (https://arxiv.org/abs/2510.00643). The arguments above are self-contained; they do not invoke that paper's convergence theorems.

The scope remains specific:

- The sharp step is used for the smooth squared-norm bound; the generalized-smooth bound uses an exact downlink and a constant-within-run radius. Its stationarity criterion is a first moment, not a second moment.
- Both vanishing stochastic rates use horizon-dependent $\mu = 1 - T^{-1/2}$. The proofs do not establish a vanishing rate for fixed $\mu = 0.95$ with fixed-variance gradients and the experimental schedules.
- The oracle is exact. A finite Newton–Schulz approximation needs an additional error analysis.
- The arbitrary-pair layer assumptions from the article are preserved. They imply separability across blocks. This note does not extend the theorems to arbitrarily coupled neural-network layers.
- The subsequent Lean development verifies the full stochastic algorithm, including the paper's round order, with the Lyapunov constants in the shared article appendix. The constants in this earlier note are separate and are not claimed to be the constants checked by Lean.